Outline: Two can system modeling and simulation

1. Part 1: Overview and goals of the lab
2. Part 2: Modeling the one-can system
3. Part 3: Steady-state versus dynamic emptying
4. Part 4: Finding K from experiments
5. Part 5: The two-can system model and simulation
6. Part 6: Summary + Pre-lab and Lab Exercises
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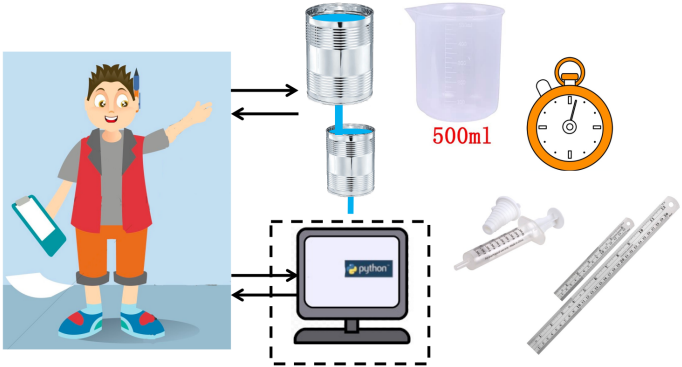
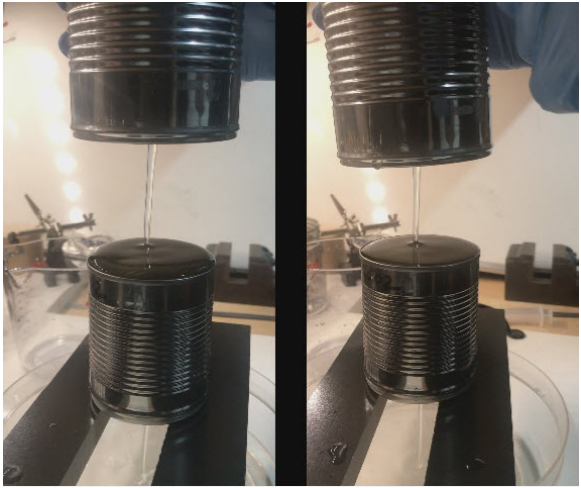
Part 1: Overview and goals of the lab



We cascade two cans so that one empties into the other, which then empties to a reservoir. We develop models to guide physical experimentation and simulation studies for predicting performance.

Spilling and filling



You see tank emptying problems beginning in physics classes

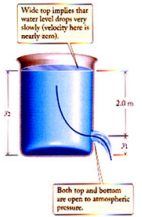


FIGURE 18.31 Streamline for a parcel of water flowing out of a tank.

EXAMPLE 12 A water tank has a (small) hole near its bottom at a depth of 2.0 m from the top surface (see Fig. 18.31). What is the speed of the stream of water emerging from the hole?

SOLUTION: Qualitatively, one of the streamlines for the water flowing out of the tank will look as shown in Fig. 18.31. Since the hole is small, the water level at the top of the tank drops only very slowly; we can therefore take $v_1 = 0$ in Eq. (18.25). Furthermore, the pressures at the top and in the emerging stream of water are the same; both are equal to the atmospheric pressure p_0. Thus, $p_1 = p_2 = p_0$. With this, Eq. (18.25) becomes

$$\tfrac{1}{2}\rho v_2^2 + \rho g y_2 + p_0 = \rho g y_1 + p_0 \qquad (18.28)$$

We can cancel the terms p_0, and we can move the term $\rho g y_2$ to the right side of the equation:

$$\tfrac{1}{2}\rho v_2^2 = \rho g y_1 - \rho g y_2 \qquad (18.29)$$

From Physics for Engineers and Scientists, Ohanian and Markert, 3rd ed

If we divide both sides by $\frac{1}{2}\rho$ and extract the square root of both sides, we find

$$v_2 = \sqrt{2g(y_1 - y_2)} \qquad (18.30)$$

With $y_1 - y_2 = 2.0$ m, the speed is

$$v_2 = \sqrt{2 \times 9.81\ \text{m/s}^2 \times 2.0\ \text{m}} = 6.3\ \text{m/s}$$

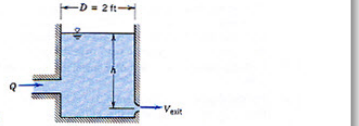
COMMENT: According to Eq. (18.30), the speed of the emerging water is exactly what it would be if the water were to fall freely through a height $y_1 - y_2$ [compare Eq. (2.29)]. This result is called **Torricelli's theorem**. It merely expresses conservation of energy: when a drop of water flows out at the bottom, the loss of potential energy of the water in the tank is equivalent to the removal of a drop of water from the top; the conversion of this potential energy into kinetic energy will give the drop the speed of free fall. We have already relied on this energy argument in Chapter 8, where we calculated the speed of water flowing out of a pipe in a hydroelectric storage plant.

From Physics for Engineers and Scientists, Ohanian and Markert, 3rd ed

Then you study these problems in more detail in fluid mechanics

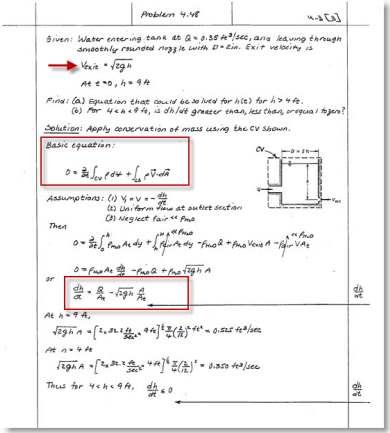
From: Introduction to Fluid Mechanics, Fox and McDonald, 4th edition

4.48 Water enters the tank through a pipe at rate $Q = 0.35$ ft³/sec. Water also leaves the tank through a smoothly rounded nozzle of 2 in. diameter. The exit velocity through this hole, which depends on the height, h, of the water level above the hole, is $V_{exit} = \sqrt{2gh}$. At $t = 0$, $h = 9$ ft. Derive an equation that could be solved for $h(t)$ when $h > 4$ ft. For $4 < h < 9$ ft, is dh/dt greater than, less than, or equal to zero?



P4.48

In this example, control volume equations are used to derive an ODE in terms of the height, h, of fluid in this tank.



Problem 4.48

Given: Water entering tank at $Q = 0.35$ ft³/sec, and leaving through smoothly rounded nozzle with $D = 2$ in. Exit velocity is

$V_{exit} = \sqrt{2gh}$

at $t = 0$, $h = 9$ ft

Find: (a) Equation that could be solved for $h(t)$ for $h > 4$ ft.
(b) For $4 < h < 9$ ft, is dh/dt greater than, less than, or equal to zero?

Solution: Apply conservation of mass using the CV shown.

Basic equation:

$$0 = \frac{\partial}{\partial t}\int_{CV} \rho \, d\forall + \int_{CS} \rho \vec{V} \cdot d\vec{A}$$

Assumptions: (1) $V_1 = V = -\frac{dh}{dt}$
(2) Uniform flow at outlet section
(3) Neglect fair << ρ_{H_2O}

Then

$$0 = \frac{\partial}{\partial t}\int_0^h \rho_{H_2O} A_t \, dy + \int_h^{h+\Delta h} \rho_{fair} A_t \, dy - \rho_{H_2O} Q + \rho_{H_2O} V_{exit} A - \rho_{fair}^{<<\rho_{H_2O}} V A_t$$

$$0 = \rho_{H_2O} A_t \frac{dh}{dt} - \rho_{H_2O} Q + \rho_{H_2O} \sqrt{2gh}\, A$$

or

$$\frac{dh}{dt} = \frac{Q}{A_t} - \sqrt{2gh}\,\frac{A}{A_t}$$

At $h = 9$ ft,

$$\sqrt{2gh}\, A = \left[2 \times 32.2 \frac{ft}{sec^2} \times 9\,ft\right]^{\frac{1}{2}} \frac{\pi}{4}\left(\frac{2}{12}\right)^2 ft^2 = 0.255\ ft^3/sec$$

At $h = 4$ ft

$$\sqrt{2gh}\, A = \left[2 \times 32.2 \frac{ft}{sec^2} \times 4\,ft\right]^{\frac{1}{2}} \frac{\pi}{4}\left(\frac{2}{12}\right)^2 = 0.320\ ft^3/sec$$

Thus for $4 < h < 9$ ft, $\frac{dh}{dt} < 0$

Summary of lab goals and objectives

Goals of this lab	How we'll achieve this goal
Practice modeling a dynamic system	Practical hydraulic system modeling, orifice flow
Learn about sensors	Only basic instruments required + observation
Learn to use sensors	Practice making basic measurements on purpose
Learn about and apply methods and principles for measurement of signals	No signals, just watching
Experiment design & practice	Model-based experiment design
Answer specific question(s)	Can I reliably predict filling height? Can I exactly fill the bottom can through model prediction?

Part 2: Modeling the one-can system

In dynamic systems modeling, we learn to model by identifying distinct basic 'elements' and use physical laws to understand how they are interconnected.

- Fluid potential energy storage in a can (or tank) is modeled as a **C** element
  - quasi-static assumption
  - constant-area tank
  - pressure-volume $(P - V)$ constitutive relation
- Orifice flow induces energy dissipation modeled by using an **R** element
  - Steady-flow Bernoulli model
  - simplifying assumptions
  - pressure-flow $(P - Q)$ constitutive relation

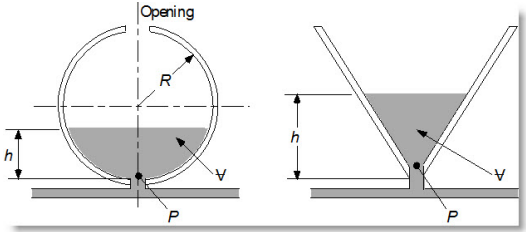
The hydraulic can or tank element

A *can* only stores potential hydraulic energy (if we can ignore inertial effects), and we show that this is a capacitive element by recalling that,

$$P = \rho g h = \frac{\rho g}{A} V = \frac{1}{C} V$$

where the can is assumed to have constant cross-sectional area, so C is a hydraulic capacitance, $C = A/(\rho g)$.

If the can or tank does not have constant cross-sectional area, as is common in many applications as illustrated below, then the constitutive law would not be linear.



It may also be the case that you cannot ignore inertial effects of the fluid. What would the tank look like in such a case?

An Orifice/Valve as an **R** Element

You can show that orifice flowrate, Q, can be directly related to pressure drop across the orifice, P, by,

$$Q = C_c A_o \sqrt{2P/\rho}$$

where C_c is a discharge coefficient, A_o the orifice area.

We identify the flowrate as a *flow* variable and pressure as an *effort* variable, the product being the *dissipated power*, $\mathcal{P} = P \cdot Q$. Since the orifice involves dissipative processes, we model the orifice as a resistive (**R**) element.

Orifice flow relation

The flowrate relation derived is used extensively for orifice flow predictions in a wide range of applications. For example, it is commonly applied when predicting hydraulic valve flows.

If the pressure can reverse sign, then the relation may need to be generalized to account for sign by using,

$$Q = C_c A_o \mathrm{sgn}(P)\sqrt{2|P|/\rho}$$

where the sgn() function provides sign information and the absolute value of pressure is taken when used in computation.

Note: detailed discussion of the orifice flow rate model can be found in the appendix.

One can dynamic system model

We assemble the elements together using the *mass continuity equation* where,

$$\dot{m}_{stored} = \sum \dot{m}_{in} - \sum \dot{m}_{out}$$

where assuming incompressible flow,

$$\rho \dot{V} = \sum \rho Q_{in} - \sum \rho Q_{out}$$

For the one can system,

$$\dot{V} = \sum Q_{in} - \sum Q_{out}$$

where V, the volume stored in the can, is the system state.

$\Rightarrow$ Recognize that this equation cannot be solved until the terms on the right-hand side are expressed as functions of states or inputs.

One can dynamic system model state equation

To derive the state equation for the one-can system, use the constitutive relations discussed earlier. First,

$$\frac{dV}{dt} = Q_{in} - Q_{out}$$

where

$$Q_{out} = C_c A_o \sqrt{2P/\rho}.$$

But,

$$P = V/C$$

Now,

$$\frac{dV}{dt} = Q_{in} - C_c A_o \sqrt{2V/\rho C}$$

which can be simplified to the state equation,

$$\frac{dV}{dt} = Q_{in} - K\sqrt{V}$$

where K is a constant for a given can.

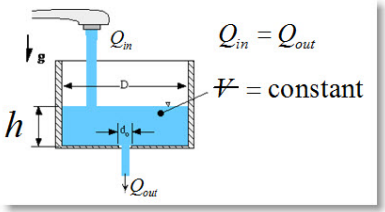
This equation is in the form of a 1st order state equation.

Steady-state versus dynamic emptying

A **steady-state condition** occurs when Q_{in} is equal to Q_{out} so,

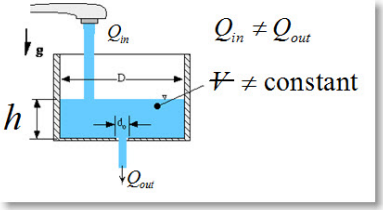
$$\frac{dV}{dt} = Q_{in} - Q_{out} = 0.$$

The system is said to be in equilibrium or steady-state and volume will be constant.



A **dynamic state** occurs when $Q_{in} \neq Q_{out}$, so the tank dynamically empties and/or fills over time, with volume a dynamic state determined by,

$$\frac{dV}{dt} = Q_{in} - K\sqrt{V}$$



Part 3: Steady-state versus dynamic emptying

Steady-state: *In general, setting $dx/dt = 0$ for all your state equations gives n algebraic equations that can be solved for the equilibrium states.*

In this case,

$$\frac{dV}{dt} = 0.$$

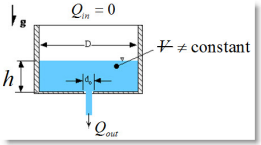
gives $Q_{in} = Q_{out}$, which can be solved for V_e, the steady-state or equilibrium volume.

Q. Can you solve for V_e?

Dynamic state: The special case $Q_{in} = 0$ with known V_o is the can emptying problem,

$$\frac{dV}{dt} = -K\sqrt{V}$$

which can be solved for $V(t)$ in closed form, given we know K.



Q. Can you find $V(t)$?

Part 4: Finding K from experiments

The emptying can model can be used to design an experiment to determine the unknown parameter, K.

The equation,

$$\frac{dV}{dt} = -K\sqrt{V}$$

is a nonlinear ordinary differential equation that can be integrated from the form,

$$\int \frac{dV}{\sqrt{V}} = -\int K dt$$

An 'emptying experiment' begins with an initial volume, V_o, that empties in time, T_e, as expressed through the integration,

$$\int_{V_o}^{0} \frac{dV}{\sqrt{V}} = -\int_{0}^{T_e} K dt$$

Model analysis

Integration of the equation,

$$\int_{V_o}^{0} \frac{dV}{\sqrt{V}} = -\int_{0}^{T_e} K dt$$

yields,

$$\int_{V_o}^{0} \frac{dV}{\sqrt{V}} = 2V^{1/2}|_{V_o}^{0} = -\int_{0}^{T_e} K dt = -Kt|_{0}^{T_e}$$

giving,

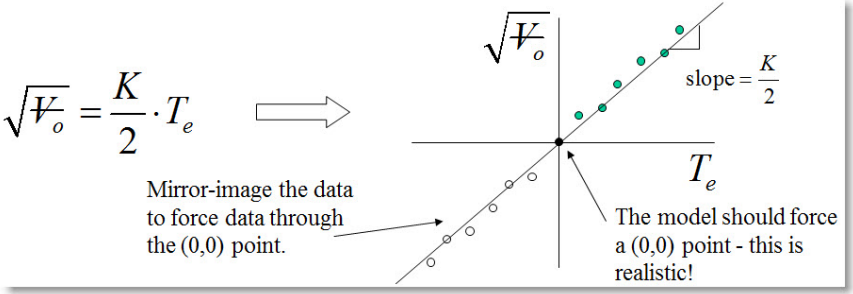
$$2\sqrt{V_o} = KT_e.$$

This relation suggests that measurements of initial volume and time-to-empty can be used to estimate the parameter K.

We use this as an example of *model-based experiment design*.

Data collection and analysis

The relation, $2\sqrt{V_o} = KT_e$, suggests there is a linear relation between initial volume and time-to-empty.



Q. Do you think it is correct to force the trend line in this relation to go through (0,0)?

Determining K from a steady-state experiment

The other way to find a K value is to use a steady-state experiment.

1. Describe what measurements would be needed to find K.
2. Explain how the physical conditions are different from the those in a *dynamic* can emptying experiment.
3. Is there a preference in choosing one of the experimental approaches described over the other?

Part 5: The two-can system model and simulation

The state equations for the two can system are found by simply applying continuity on the 'second' can.

You should be able to show that the state equations for the volumes in cans 1 (top) and 2 (bottom) are, respectively,

$$\begin{aligned} \dot{V}_1 &= -K_1\sqrt{V_1} \\ \dot{V}_2 &= +K_1\sqrt{V_1} - K_2\sqrt{V_2} \end{aligned}$$

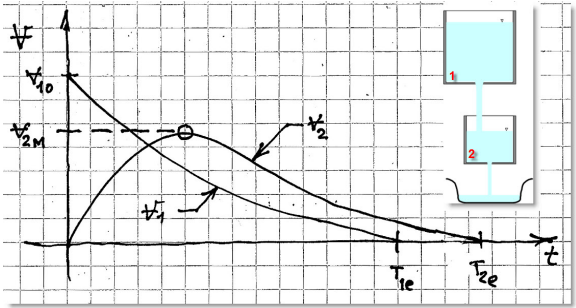
where the K values are distinct for each can. Note that there is no input flow to top can, but it can be readily added.

These equations can not be integrated analytically, but can be solved using numerical simulation.

Two-can system dynamics

Consider the case where the top can (1) is filled to some level and the bottom can (2) is initially empty. When released, flow from the top can will begin to fill the bottom can, which then empties (to a reservoir).

The dynamic response of each can volume should follow trends similar to those shown below.



System equations in Python

The two-can system ODEs can be formatted in a Python function as shown to the right.

This function computes the the flow rates as intermediate variables, `Q1` and `Q2`.

These flows can then be used to define the ODEs within a `np.array()` element as shown.

The flows can also be passed in the output `y`.

```python
def two_can(x, t, K1, K2):
    V1, V2 = x[0], x[1]
    if V1 <= 0.0:
        Q1 = 0
    else:
        Q1 = K1*np.sqrt(abs(V1))

    if V2 <= 0.0:
        Q2 = 0
    else:
        Q2 = K2*np.sqrt(abs(V2))

    # specify outputs
    y = [Q1, Q2]

    return np.array([-Q1, +Q1-Q2]), y
```

Extracting the outputs using the system function

For the two-can function file, the output flows can be computed after the simulation has been run.

Since the solution is found in terms of 'states', any variables of the system can be computed in terms of the states.

We know this is the case for the flow rates.

The method for computing the flows is as shown to the right.

```
sol = rk.rk4fixed(two_can, x0, t, args=(K1, K2))
V1 = sol[:,0]
V2 = sol[:,1]
# compute outputs
Q1, Q2 = [], []
for i in range(len(t)):
    _, y = two_can(sol[i,:], t[i], K1, K2)
    Q1.append(y[0])
    Q2.append(y[1])
```

Experimental testing and evaluation of the two-can model

- Testing a one-can model. Given an experimentally determined, K, repeated tests can be run to determine if time-to-empty is accurately predicted. To fully assess accuracy of the model, the volume needs to be measured over time and compared to simulation predictions.
- Testing a two-can model. One quick test is to begin with top can at a given volume, bottom can empty, and estimate peak level reached in the bottom can and time-to-empty. This gives a first assessment of accuracy of the combined system model. As for the one-can system, measuring volume over time would give a more accurate assessment.

Part 6: Summary

- The one and two-can systems rely on physical modeling of hydraulic system elements, and introduce the application of state space modeling.
- The system model is described by two nonlinear differential equations in state space form.
- To make predictions about the two volume states of this system, it is necessary to solve the equations using a numerical simulation program.
- In addition, each can requires its own 'K' value which must be experimentally determined.

Overview of the laboratory exercises

1. Become familiar with laboratory setup and basic tools
2. Run experiments to find the empirical flow coefficent, K, for each can
3. Review simulation of two-can system
4. TA to discuss model validation methods
5. TA will evaluate your model results through two-can validation tests (fill or spill)

Pre-lab discussion

1. Find theoretical K coefficients for typical laboratory cans
2. Run a typical two-can simulation in Python using theoretical K values and given can dimensions
3. Think through the lab procedure for finding K values from experiments

Appendix: Theoretical orifice flow model

Begin by using mass continuity and the steady Bernoulli equation to derive a relation for the velocity of the fluid exiting at control surface 2 (see right).

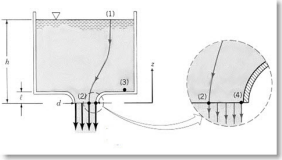
In this case,

$$0 = \frac{\partial}{\partial t}\int_{CV} \rho dV + \int_{CS} \rho \vec{v} \cdot d\vec{A}$$

$$\frac{P_1}{\rho} + \frac{v_1^2}{2} + gh_1 = \frac{P_2}{\rho} + \frac{v_2^2}{2} + gh_2$$

to show that v_2 is given by:

$$v_2 = \sqrt{\frac{2gh}{[1-(A_2/A_1)^2]}}$$



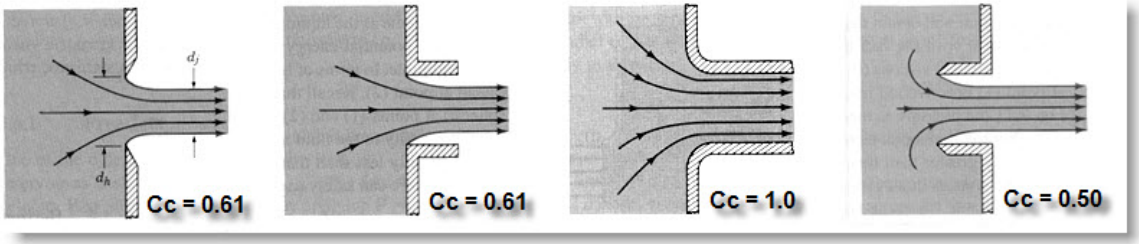
What is A_2? Approximate by using the jet area. If there is an orifice where the jet area is not equal to the orifice area, we use a contraction coefficient.

Orifice contraction coefficients

The contraction coefficient is defined by,

$$C_c = \frac{A_{jet}}{A_o}$$

Some common experimentally determined values:



Velocity coefficient

Using the contraction coefficient, our model for the exiting mass flowrate is,

$$\dot{m} = \rho Q = \rho A_{\text{jet}} v = \rho C_c A_o C_v v_{\text{ideal}}$$

where we've used the velocity coefficient,

$$C_v = \frac{v_{\text{actual}}}{v_{\text{ideal}}} = \text{velocity coefficient}$$

which accounts for friction effects.

Orifice flowrate model

Use the ideal velocity from the Bernoulli equation,

$$v_{\text{ideal}} = v_2 = \sqrt{\frac{2gh}{[1-(A_2/A_1)^2]}}$$

and gauge pressure, P, at the bottom of the can so the flowrate from the orifice is,

$$Q = \frac{\dot{m}}{\rho} = \frac{C_c C_v A_o}{[1-C_c^2(A_2/A_1)^2]}\sqrt{\frac{2P}{\rho}} = K_o A_o \sqrt{\frac{2P}{\rho}}$$

where ρ is assumed constant and,

$$K_o = \frac{C_c C_v}{[1-C_c^2(A_2/A_1)^2]}$$

Ideal orifice flow constant, K_{ideal}

In the derived expression for K_o, assume ideal contraction and velocity coefficients, $C_c = 1$ and $C_v = 1$ so,

$$K_{o,\text{ideal}} = \frac{1}{\sqrt{1 - (A_{\text{orifice}}/A_{\text{can}})^2}}$$

and the ideal case flowrate is,

$$Q_{\text{ideal}} = K_{o,\text{ideal}} A_o \sqrt{\frac{2}{\rho}} \sqrt{\frac{V}{C}} = K_{o,\text{ideal}} A_o \sqrt{\frac{2}{\rho} \frac{\rho g}{A_{\text{can}}}} \sqrt{V} = K_{\text{ideal}} \sqrt{V}$$

where the ideal can coefficient is defined by,

$$K_{\text{ideal}} = K_{o,\text{ideal}} A_o \sqrt{2g/A_{\text{can}}},$$

a value calculated from easily measured dimensions.

Example K_{ideal} calculations

Example calculation of ideal flow coefficient calculations given in different units are shown below.

$$D_{can} := 3 \cdot in \qquad D_o := \frac{1}{8} \cdot in$$

$$A_{can} := \frac{\pi \cdot D_{can}^{\,2}}{4} \qquad A_o := \frac{\pi \cdot D_o^{\,2}}{4}$$

$$K_{oideal} := \frac{1}{\sqrt{1 - \left(\frac{A_o}{A_{can}}\right)^2}} \qquad K_{ideal} := K_{oideal} \cdot A_o \cdot \sqrt{\frac{2 \cdot g}{A_{can}}}$$

$$K_{oideal} = 1.000001507$$

$$K_{ideal} = 0.128 \frac{\sqrt{in^3}}{s} \qquad K_{ideal} = 5.192 \times 10^{-4} \frac{\sqrt{m^3}}{s} \qquad K_{ideal} = 0.128 \frac{\sqrt{in^3}}{s}$$